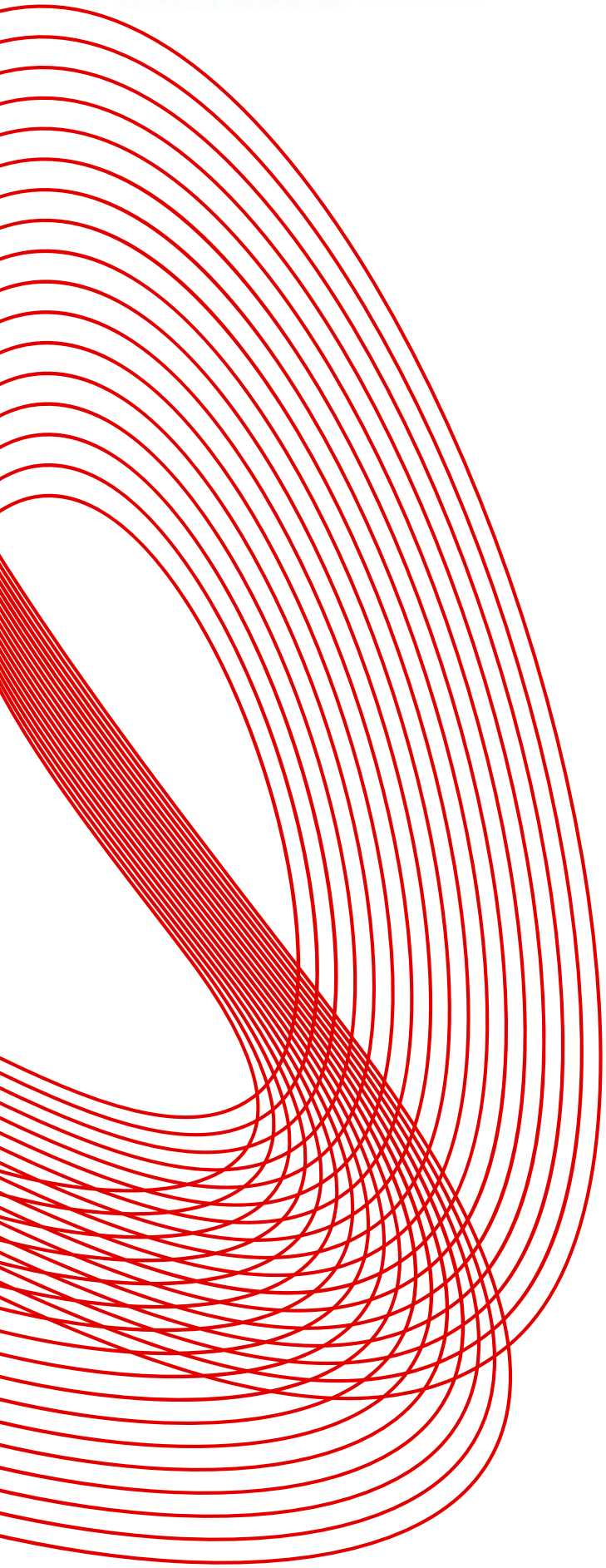
INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES STRASBOURG

S9



S-IoT Project Report :

Drone swarm

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Specifications

Objective

The purpose of this project is to fly a swarm of UAVs (2 to 3 UAVs) and try to lift off a payload using a net.

Functions

- UWB precise location system
- PID Controller for drone stabilization
- UWB Communication for high frequency refreshing data
- Go back and forth keeping the drone formation (following a leader ?)

Deliverables

- Report with all progress, advancements and explanation of technical knowledge
- Functional UWB code
- Methodology and results of calibration of UWB sensors
- Bibliographical Studies
- Video of drone demo
- Functional PID Corrector
- Functional ESP32 code implementation using Multi-threading

Bibliographical Studies

0.1 UWB

0.1.1 Definitions

Ultra-Wideband (UWB) is a technology using radio frequency to exchange data created in 1880. Communication is not the only purpose, it can also be used to locate; it is only possible by using different module, those algorithms and technique will be treated in the next part (**??**). In the majority of cases, UWB is deployed for indoor purpose, otherwise, it can be used for outdoor purpose; but it is less shown in scientific papers.

0.1.2 Location algorithm

In order to use UWB as a location sensor. We have to use specific algorithm; in the following subsection, we will explain and show 2 different location algorithms. Firstly, two-way ranging (TWR) and secondly double sided two-way ranging (DS-TWR).Finally, we will describe TDOA, the most advanced algorithms.

Two way ranging (TWR)

The Two-Way Ranging (TWR) algorithm is the easiest and the fastest to implement in a UWB system. In order to explain this algorithm, we will use 2 modules called UWB_1 and UWB_2.

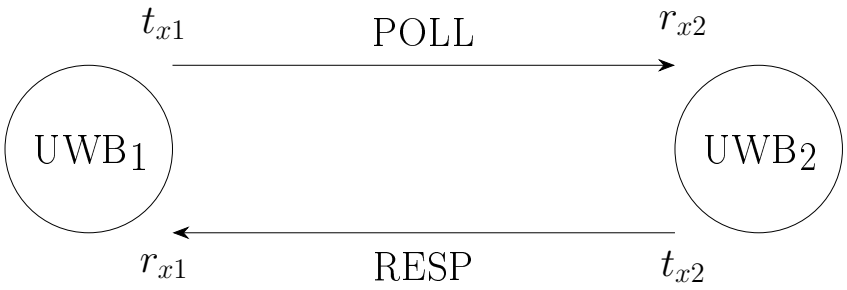


Figure 1: TWR explanation diagram

The algorithm works in 4 steps :

1. UWB_1 sends a **POLL** to UWB_2. The time of emission t_{x1} is memorized by UWB_1.
2. UWB_2 receives the **POLL** and memorizes the time of arrival r_{x2}.
3. UWB_2 processes the **POLL**, and precisely plans the time of emission t_{x2} for the **RESP**. It contains (t_{x2}, r_{x2}).
4. UWB_1 receives the **RESP** and memorizes the r_{x1}.

With all those parameters, we can deduce those variables :

- Total time : $t_{tot} = r_{x1} - t_{x1}$
- Processing time of UWB_2 : $t_{proc} = r_{x2} - t_{x2}$
- Travel time : $t_{travel} = t_{tot} - t_{proc}$

The signal travels back and forth so it did twice the distance (d), and travel speed of a radio signal is equivalent to light speed (c).
We can deduce this equation :

$$\boxed{\frac{t_{travel} \times c}{2} = \frac{(t_{tot} - t_{proc}) \times c}{2} = \frac{((r_{x1} - t_{x1}) - (r_{x2} - t_{x2})) \times c}{2}} \quad (1)$$

We can also determine the induced error of this algorithm.
After calculation (0.2.2), we obtain this equation:

$$\epsilon \approx \frac{t_{proc}(e_a - e_b)}{2} \quad (2)$$

As we can observe, the Time of processing is directly linked to the error. The t_{proc} is around 3,000,000 nanoseconds so, it is inducing a high error.

Double Sided Two way ranging (DS-TWR)

In the TWR algorithm, the main inconvenient is the error and misprecision. To erase that error, the DS-TWR has been thinked. This algorithm is also using UWB_1 and UWB_2.

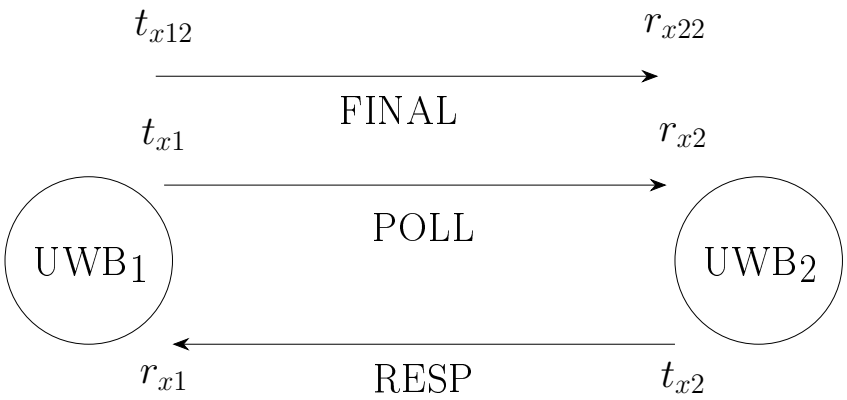


Figure 2: DS-TWR explanation diagram

The algorithm works in 6 steps :

1. UWB_1 sends a **POLL** to UWB_2. The time of emission t_{x1} is memorized by UWB_1.
2. UWB_2 receives the **POLL** and memorizes the time of arrival r_{x2}.
3. UWB_2 processes the **POLL** and precisely plans the time of emission t_{x2} of the **RESP**, so that the reply delay $D_b = t_{x2} - r_{x2}$ is known.
4. UWB_1 receives the **RESP** and memorizes the time of arrival r_{x1}.
5. UWB_1 processes the **RESP** and precisely plans the time of emission t_{x12} of the **FINAL**, so that the reply delay $D_a = t_{x12} - r_{x1}$ is known. The **FINAL** contains (t_{x1}, r_{x1}, t_{x12}).
6. UWB_2 receives the **FINAL** and memorizes the time of arrival r_{x22}.

With all those parameters, we compute four time intervals :

- Response time for $UWB_1 : t_1 = r_{x1} - t_{x1}$
- Response time for $UWB_2 : t_2 = r_{x22} - t_{x2}$
- Processing time of $UWB_1 : t_{proc_1} = t_{x12} - r_{x1}$

- Processing time of $UWB_2 : t_{proc_2} = r_{x2} - t_{x2}$

We can deduce the travel time:

$$\boxed{t_{travel} = \frac{t_1 \times t_2 - t_{proc_1} \times t_{proc_2}}{t_1 + t_2 + t_{proc_1} + t_{proc_2}}} \tag{3}$$

And then find distance :

$$d = c \times t_{travel} \tag{4}$$

Now let's compare the induced error. After calculation (0.2.2), we obtain this equation :

$$\epsilon \approx T_f \times \frac{e_a + e_b}{2} \tag{5}$$

As we can see, the error is not multiplie by t_{proc}, so the error is reduced.
With this algorithm, we are note dependant of the processor any more. That the algorithm currently use in our project.

0.1.3 Timing algorithm

When working on swarm drone, there is not only two UWB modules. There is multiple UWB trying to communicate. One weakness of the UWB is it sensibility to noise and interference. The way should be clear before emiting otehrwise that create colision and damaged the singal and corrupt distance estimation. For this purpose, we need to look on how to divide time to avoid collision, in our specifif case, we are working with 3 to 4 drone.

ALOHA

This method is named after the first ALOHAnet system, which introduced it in 1977. It provides a simple approach to channel access control without requiring any time synchronization. Instead of planning transmissions, a module sends its data packet immediately, regardless of whether the medium is currently busy or free. The primary advantage of ALOHA lies in its ease of implementation and full decentralization, allowing independent nodes or drones to join the network dynamically without prior initialization. However, its major drawback is its susceptibility to collisions caused by uncoordinated simultaneous transmissions. As demonstrated in foundational studies, the maximum theoretical throughput (channel capacity) of ALOHA is limited to approximately 18.4% (ref du papier). This low efficiency makes it unsuitable for high-frequency localization and critical communication applications.

Time-division multiple access (TDMA)

This method has for idea to divide a fixed period of time in trame, and cut it in different tranches. Each tranche is for each drone. Those trames and tranches are fixed in the hardware and are stricly respected.

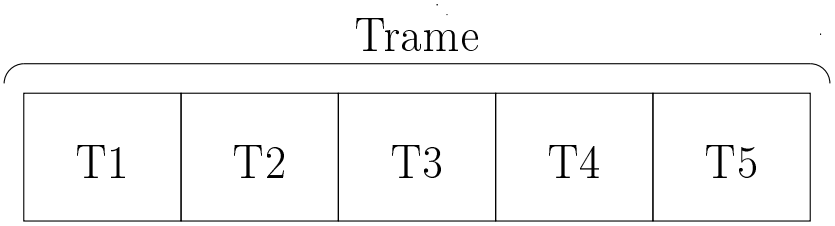


Figure 3: TDMA explanation diagram

For exemple if we have 3 drone, trame of UWB_1 when is looking for distance in DS-TWR algorithm is divided by 5.

1. T1: UWB_1 send the **POLL** to UWB_2 and UWB_3
2. T2: UWB_2 send the **RESP** to UWB_1
3. T3: UWB_1 send the **FINAL** to UWB_2
4. T4: UWB_3 send the **RESP** to UWB_1
5. T5: UWB_1 send the **FINAL** to UWB_3

This method has an advantage: there is no collision at all due to strict time period. And also permit to shut down the module when we do not need it. But there is inconvenient, firstly if we hade one more drone, we have to add two other tranches, when we reach 10 drone, there is no time anymore to get information at a high frequency. And, it needs a synchronization to make all clock working at same speed and avoid collision perfectly.

Slotted ALOHA

This method combines the features of both the ALOHA and TDMA protocols. Time is divided into tranches, and UWB modules are only allowed to start their transmissions at the beginning of a slot boundary. If a module has data ready to transmit, but a slot has already started, it must wait for the next available slot.

This approach preserves the flexibility of ALOHA while doubling the maximum theoretical throughput to approximately 36.8% (ref du papier). However, it still remains less deterministic and efficient than TDMA due to potential collisions when multiple nodes choose the same slot.

0.2 Measure processing

In most cases, date measured by sensors are unclear. It could be a sensor offset, it can also come from crosstalks. Or other, exterior interactions. In order, to obtain clean measurements, and to be able to use them for controlling or any other purpose. We have to develop some filter; in this section, we will describe three filters used for cleaning the UWB signal. Firstly, we will explain the polynomial filter, and secondly, the Kalman filter, then finally the extended Kalman filters

0.2.1 Polynomial Filter

colorblue

0.2.2 Extended Kalman Filter for relative localisation

In the context of navigating systems in noisy environments with sensors, the Kalman filter is the most relevant algorithm. As Welch and Bishop [1] point out, the Kalman filter is notable for its ability to "support estimations of past, present, and even future states [...] even when the precise nature of the modeled system is unknown.". In our study case, we deal with a drone swarm communicating using the UWB communication protocol, which gives us the relative distance between each pair of drones by using the time of flight of a radio wave. This measure is non-linear (Euclidian distance calculated with a square root) which implies that an extended version of the filter should be used : it resorts to a typical technique of linearization around the current state using a Jacobian matrix. We are also considering an anchor-free system which enhances the challenge of this project. In this section, we will first define the theory of the EKF (Extended Kalman Filter) and then, we will study its application to our UWB drone swarm system.

Theory of the Kalman filter

Origin of the theory In his pioneering article of 1960 [2], Rudolf Kalman describes state estimation as a recursive process in which at each time step, the current estimation and the uncertainty are computed using only the past estimate and the newly available measure. When assuming a linear dynamical system and Gaussian noise, the Kalman filter minimizes the mean-square estimation error : in other words, the best possible estimate given the available measurements. Even when these assumptions do not hold exactly, the filter remains the best linear estimator achievable (known as the Best Linear Unbiased Estimator, or BLUE). This optimality is what established the Kalman filter as the standard approach for state estimation in noisy dynamic systems. While Kalman filter's optimality holds for linear dynamical systems, we have in our case, as mentioned earlier, a non-linearity with the use of the Euclidian distance. This is why we require an extended version of the filter that can handle the non-linearity, even at the cost of losing the strict optimality.

Formalism of the discrete Kalman Filter Let's note $x_k \in \mathbb{R}_n$ the current state of a discrete system governed by the following linear equation :

$$\mathbf{x}_k = \mathbf{F}_k\mathbf{x}_{k-1} + \mathbf{B}_k\mathbf{u}_k + \mathbf{w}_k$$

In which we have :

- F_k is the state transition matrix, linking the previous state x_{k-1} to the current state x_k
- B_k is the control-input matrix, linking the control vector u_k to the current state x_k
- w_k is the process noise which is assumed to be Gaussian and centered, with a covariance matrix Q_k

At each time k, the measurement z_k is written as follows :

$$\mathbf{z}_k = \mathbf{H}_k\mathbf{x}_k + \mathbf{v}_k$$

With :

- H_k is the observation matrix
- w_k is the observation noise which is assumed to be Gaussian and centered, with a covariance matrix R_k

We can divide the Kalman filter process into 2 phases : the "prediction" phase and the "update" phase. The first one does an estimate of the state at the next time step by using the estimate of the current time step and the error covariance. The second phase can be thought of as a feedback : it includes a new measurement in the current state to improve its estimate.

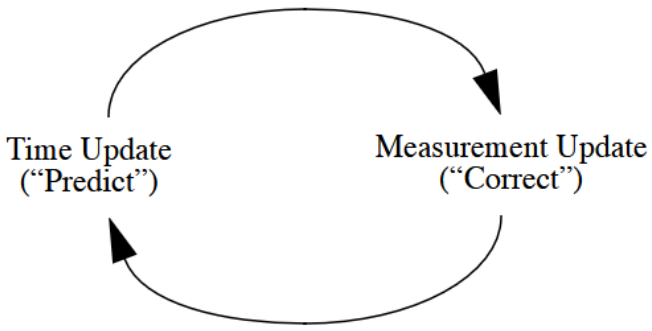


Figure 4: Ongoing discrete Kalman filter cycle (Welch and Bishop (2006, p. 5) [1])

Extended Kalman filter

As mentioned earlier, the extended Kalman filter is used to deal with a non-linear state process defined as diffentiable function f. To do so, it essentially linearizes the nonlinear function around the current estimate by computing the Jacobian matrices of the nonlinear functions f and h in the following equations (with the same notations as before) :

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{w}_k)$$
$$\mathbf{z}_k = h(\mathbf{x}_k, \mathbf{v}_k)$$

Calculations

TWR Error calculation :

Let's set:

- True time of flight : T_f
- True time of process : t_{proc}
- Clock drift errors : e_a and e_b

The measured times are:

$$R_a = (2T_f + t_{proc})(1 + e_a) \text{ and } t_{proc_b} = t_{proc}(1 + e_b)$$

The calculated time of flight is:

$$\hat{T}_f = \frac{R_a - t_{proc_b}}{2}$$

By replacing R_a and t_{proc_b}, we obtain :

$$\hat{T}_f = \frac{2T_f(1 + e_a) + t_{proc}(1 + e_a) - t_{proc}(1 + e_b)}{2}$$

By simplifying, we obtain :

$$\hat{T}_f = T_f(1 + e_a) + \frac{t_{proc}(e_a - e_b)}{2}$$

The error is $\hat{T}_f - T_f$, so:

$$Error = T_f \times e_a + \frac{t_{proc}(e_a - e_b)}{2}$$

The time of flight is about 33 nanoseconds, however t_{proc} is around $3,000,000$ns.
So we can ignore $T_f \times e_a$ cause $T_f \ll t_{proc}$, and we obtain :

$$Error \approx \frac{t_{proc}(e_a - e_b)}{2} \tag{6}$$

DS-TWR Error calculation :

Let's set:

- True time of flight : T_f
- True times of process : t_{pa} and t_{pb}
- Clock drift errors : e_a and e_b

The measured times are:

$$R_a = (2T_f + t_{pb})(1 + e_a), \quad D_a = t_{pa}(1 + e_a)$$

$$R_b = (2T_f + t_{pa})(1 + e_b), \quad D_b = t_{pb}(1 + e_b)$$

The calculated time of flight is:

$$\hat{T}_f = \frac{R_a R_b - D_a D_b}{R_a + R_b + D_a + D_b}$$

By replacing the measured times, we obtain :

$$\hat{T}_f = \frac{(1+e_a)(1+e_b) \times 2T_f(2T_f + t_{pa} + t_{pb})}{(2+e_a+e_b)(2T_f + t_{pa} + t_{pb})}$$

By simplifying, we obtain :

$$\hat{T}_f = \frac{2T_f(1+e_a)(1+e_b)}{2+e_a+e_b} \approx T_f\left(1 + \frac{e_a+e_b}{2}\right)$$

The error is $\hat{T}_f - T_f$, so:

$$Error \approx T_f \times \frac{e_a+e_b}{2} \tag{7}$$

References

[1] G. Welch and G. Bishop, "An Introduction to the Kalman Filter," University of North Carolina at Chapel Hill, Tech. Rep. TR 95-041, 2006.

[2] R. E. Kalman, "A New Approach to Linear Filtering and Prediction Problems," en, *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, Mar. 1960, ISSN: 0021-9223. DOI: 10.1115/1.3662552 Accessed: Oct. 5, 2026. [Online]. Available: https://asmedigitalcollection.asme.org/fluidsengineering/article/82/1/35/397706/A-New-Approach-to-Linear-Filtering-and-Prediction